

Summing using De Moivre's

Basic skills

Q1. a) De Moivre's theorem states

for any n $(\cos x + i \sin x)^n = \cos nx + i \sin nx$

$$\begin{aligned} \text{b) } & (\cos \theta + i \sin \theta)^4 \\ &= \cos^4 \theta + 4 \cos^3 \theta \times i \sin \theta \\ &\quad + 6 \cos^2 \theta \times i^2 \sin^2 \theta + 4 \cos \theta \times i^3 \sin^3 \theta \\ &\quad + i^4 \sin^4 \theta \end{aligned}$$

$$= \cos^4 \theta + \sin^4 \theta - 6 \cos^2 \theta \sin^2 \theta$$

$$+ 4i \cos \theta \sin \theta (\cos^2 \theta - \sin^2 \theta)$$

Real Part

Imaginary Part

$$\text{c) so } \operatorname{Re}((\cos \theta + i \sin \theta)^4)$$

$$= \cos^4 \theta + \sin^4 \theta - 6 \cos^2 \theta \sin^2 \theta$$

d) As $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$
we have

$$\cos 4\theta = \operatorname{Re}((\cos \theta + i \sin \theta)^4)$$

so

$$\cos 4\theta = \cos^4 \theta + \sin^4 \theta - 6 \cos^2 \theta \sin^2 \theta$$

e)

$$\begin{aligned} & \operatorname{Im}((\cos \theta + i \sin \theta)^4) \\ &= 4 \cos \theta \sin \theta (\cos^2 \theta - \sin^2 \theta) \end{aligned}$$

f) As $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$
we have

$$\begin{aligned} \sin 4\theta &= \operatorname{Im}((\cos \theta + i \sin \theta)^4) \\ &= 4 \cos \theta \sin \theta (\cos^2 \theta - \sin^2 \theta) \end{aligned}$$

Q 2.

a) For $\cos 6\theta$ use

$$(\cos \theta + i \sin \theta)^6 = \cos 6\theta + i \sin 6\theta$$

and

$$\begin{aligned}(\cos \theta + i \sin \theta)^6 &= (\overset{\text{cos } \theta}{c} + \overset{\text{sin } \theta}{i s})^6 \\&= c^6 + 6c^5 i s + 15c^4 i^2 s^2 \\&\quad + 20c^3 i^3 s^3 + 15c^2 i^4 s^4 + 6c i^5 s^5 \\&\quad + i^6 s^6 \\&= \boxed{c^6 - 15c^4 s^2 + 15c^2 s^4 - s^6} \quad \text{Real Part} \\&\quad + i(6c^5 s - 20c^3 s^3 + 6c s^5)\end{aligned}$$

As $\cos 6\theta$ is real part

$$\begin{aligned}\cos 6\theta &= \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta \\&\quad + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta\end{aligned}$$

b) For $\sin 5\theta$ need imaginary part

$$\text{as } (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{and } (\cos \theta + i \sin \theta)^5 = (c + i s)^5$$

$$= c^5 + 5c^4is + 10c^3i^2s^2 + 10c^2i^3s^3 + 5ci^4s^4 + i^5s^5$$

$$= c^5 - 10c^3s^2 + 5cs^4 + i(5c^4s - 10c^2s^3 + s^5)$$

Ignore Part

so

$$\sin 5\theta = 5\cos^4\theta \sin\theta - 10\cos^2\theta \sin^3\theta + \sin^5\theta$$

Competency skills

$$Q3. \quad C = 1 + \cos\theta + \dots + \cos 10\theta$$

$$S = 0 + \sin\theta + \dots + \sin 10\theta$$

$$a) \quad C + iS = 1 + \cos\theta + \dots + \cos 10\theta + 0 + i\sin\theta + \dots + i\sin 10\theta$$

$$= 1 + (\cos\theta + i\sin\theta) + \dots + (\cos 10\theta + i\sin 10\theta)$$

$$= 1 + z + z^2 + \dots + z^{10}$$

b) Geometric $a=1, r=z$ 11 terms

so
$$C + iS = \frac{1(1-z^{11})}{1-z} = \frac{1-z^{11}}{1-z}$$

c)
$$\frac{1-z^{11}}{1-z} = \frac{1-e^{11i\theta}}{1-e^{i\theta}} \times \frac{1-e^{-i\theta}}{1-e^{-i\theta}}$$

$$= \frac{1 - e^{-i\theta} - e^{11i\theta} + e^{10i\theta}}{1 - e^{i\theta} - e^{-i\theta} + e^0}$$

$$= \frac{1}{2 - (e^{i\theta} + e^{-i\theta})} (1 - e^{-i\theta} + e^{10i\theta} - e^{11i\theta})$$

$$= \frac{1}{2 - 2\cos\theta} (1 - (\cos(-\theta) + i\sin(-\theta)) + \cos 10\theta + i\sin 10\theta - (\cos 11\theta + i\sin 11\theta))$$

$$= \frac{1}{2(1-\cos\theta)} \left(1 - \cos\theta + i\sin\theta + \cos 10\theta + i\sin 10\theta - \cos 11\theta - i\sin 11\theta \right)$$

$$= \frac{1}{2(1-\cos\theta)} \left(1 - \cos\theta + \cos 10\theta - \cos 11\theta + i(\sin\theta + \sin 10\theta - \sin 11\theta) \right)$$

d) so $C = \frac{1}{2(1-\cos\theta)} (1 - \cos\theta + \cos 10\theta - \cos 11\theta)$

real part $\nearrow$

Imaginary part $\rightarrow S = \frac{1}{2(1-\cos\theta)} (\sin\theta + \sin 10\theta - \sin 11\theta)$

e) As infinite sum $\nwarrow$ sum to infinity

$$C + iS = \frac{1}{1-z} = \frac{1}{1-e^{i\theta}} \times \frac{1-e^{-i\theta}}{1-e^{-i\theta}}$$

$$= \frac{1}{2(1-\cos\theta)} \left(1 - (\cos(-\theta) + i\sin(-\theta)) \right)$$

$$= \frac{1}{2(1-\cos\theta)} (1 - \cos\theta + i\sin\theta)$$

$$\text{so } C = \frac{1 - \cos\theta}{2(1-\cos\theta)} = \frac{1}{2}$$

$$S = \frac{\sin\theta}{2(1-\cos\theta)}$$

Q 4.

$$C = 1 + \frac{1}{2} \cos\theta + \frac{1}{4} \cos(2\theta) + \dots$$

$$S = \frac{1}{2} \sin\theta + \frac{1}{2} \sin(2\theta) + \dots$$

$$\text{so } C + iS = 1 + \frac{1}{2} (\cos\theta + i\sin\theta) + \frac{1}{4} (\cos 2\theta + i\sin 2\theta) + \dots$$

De Moivre's

$$= 1 + \frac{1}{2} z + \frac{1}{4} z^2 + \dots$$

sum of infinite geometric series

$$\text{where } z = \cos\theta + i\sin\theta$$

$$= \frac{1}{1 - \frac{1}{2}z} = \frac{2}{2-z}$$

$$\text{So } C + iS = \frac{2}{2 - e^{i\theta}} = \frac{2}{2 - e^{i\theta}} \times \frac{2 - \bar{e}^{i\theta}}{2 - \bar{e}^{i\theta}}$$

$$= \frac{4 - 2e^{-i\theta}}{4 - e^{i\theta} - \bar{e}^{i\theta} + e^0}$$

$$= \frac{4 - 2(\cos(-\theta) + i\sin(-\theta))}{5 - (e^{i\theta} + \bar{e}^{i\theta})}$$

$$z + z^* = 2\cos\theta$$

$$= \frac{4 - 2\cos\theta + 2i\sin\theta}{5 - 2\cos\theta}$$

$$\text{So } C = \frac{4 - 2\cos\theta}{5 - 2\cos\theta} \quad \leftarrow \text{real part}$$

$$S = \frac{2\sin\theta}{5 - 2\cos\theta} \quad \leftarrow \text{imaginary part}$$

Q5. let

$$C = \cos\theta + \cos\left(\theta + \frac{2\pi}{n}\right) + \dots + \cos\left(\theta + \frac{2(n-1)\pi}{n}\right)$$

$$S = \sin\theta + \sin\left(\theta + \frac{2\pi}{n}\right) + \dots + \sin\left(\theta + \frac{2(n-1)\pi}{n}\right)$$

Then

$$C + iS = \cos \theta + i \sin \theta$$

$$+ \cos\left(\theta + \frac{2\pi}{n}\right) + i \sin\left(\theta + \frac{2\pi}{n}\right)$$

$$+ \dots + \cos\left(\theta + \frac{2(n-1)\pi}{n}\right) + i \sin\left(\theta + \frac{2(n-1)\pi}{n}\right)$$

Add
Arguments
is multiplying
Complex
Numbers

$$= \cos \theta + i \sin \theta$$

$$+ \left(\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right) (\cos \theta + i \sin \theta)$$

$$+ \dots + \left(\cos\left(\frac{2(n-1)\pi}{n}\right) + i \sin\left(\frac{2(n-1)\pi}{n}\right)\right) (\cos \theta + i \sin \theta)$$

$$= z + wz + \dots + w^{n-1}z$$

$$= z (1 + w + \dots + w^{n-1})$$

where $z = \cos \theta + i \sin \theta$

$$w = \cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)$$

so

$$C + iS = z \times \frac{1(1 - w^n)}{1 - w}$$

$$\text{but } w^n = \cos\left(\frac{2\pi}{n} \times n\right) + i \sin\left(\frac{2\pi}{n} \times n\right)$$

$$= \cos(2\pi) + i\sin(2\pi) = 1$$

So $C + iS = z \times \frac{1-z}{1-w} = z \times 0 = 0$

So $C = 0$ and $S = 0$

Advanced skills

Q6. a) $1 + e^{i\theta} = 2e^{i\frac{\theta}{2}} \cos\left(\frac{\theta}{2}\right)$

well $1 + e^{i\theta} = 1 + \cos\theta + i\sin\theta$

$\cos 2x$
 $= \cos^2 x - \sin^2 x$
 $= 2\cos^2 x - 1$

$$\begin{aligned} &= 1 + \left(2\cos^2\frac{\theta}{2} - 1\right) + i\left(2\cos\frac{\theta}{2}\sin\frac{\theta}{2}\right) \\ &= 2\cos^2\frac{\theta}{2} + 2i\cos\frac{\theta}{2}\sin\frac{\theta}{2} \\ &= 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right) \\ &= 2\cos\frac{\theta}{2}e^{i\frac{\theta}{2}} \\ &= 2e^{i\frac{\theta}{2}}\cos\frac{\theta}{2} \end{aligned}$$

$$b) \quad C = 1 + \binom{n}{1} \cos \theta + \binom{n}{2} \cos 2\theta + \dots + \binom{n}{n} \cos(n\theta)$$

$$S = 0 + \binom{n}{1} \sin \theta + \binom{n}{2} \sin 2\theta + \dots + \binom{n}{n} \sin(n\theta)$$

$$c) \quad C + iS = 1 + \binom{n}{1} (\cos \theta + i \sin \theta) + \binom{n}{2} (\cos 2\theta + i \sin 2\theta) + \dots + \binom{n}{n} (\cos(n\theta) + i \sin(n\theta))$$

$$= 1 + \binom{n}{1} z + \binom{n}{2} z^2 + \dots + \binom{n}{n} z^n$$

binomial expansion $\downarrow$ for $z = \cos \theta + i \sin \theta$

$$= (1 + z)^n = (1 + e^{i\theta})^n$$

$z = e^{i\theta}$

d) now by part a)

$$1 + e^{i\theta} = 2e^{i\frac{\theta}{2}} \cos \frac{\theta}{2}$$

so

$$C + iS = \left(2e^{i\frac{\theta}{2}} \cos \frac{\theta}{2} \right)^n$$

$$= 2^n \cos^n \frac{\theta}{2} e^{i\frac{n\theta}{2}}$$

$$= 2^n \cos^n \frac{\theta}{2} \left(\cos \frac{\theta^n}{2} + i \sin \frac{\theta^n}{2} \right)$$

SO $C = 2^n \cos^n \frac{\theta}{2} \cos \frac{\theta^n}{2} \leftarrow \text{real part}$

$S = 2^n \cos^n \frac{\theta}{2} \sin \frac{\theta^n}{2} \leftarrow \text{Imaginary part}$

Q7. a)

$$S = \cancel{1} + 2 \sin x + \frac{2^2 \sin 2x}{2!} + \frac{2^3 \sin 3x}{3!} + \dots$$

bit odd could remove now or later

$$C = 1 + 2 \cos x + \frac{2^2 \cos 2x}{2!} + \frac{2^3 \cos 3x}{3!} + \dots$$

$$\begin{aligned} C + iS &= 1 + i + 2 (\cos x + i \sin x) \\ &+ \frac{2^2}{2!} (\cos 2x + i \sin 2x) \\ &+ \frac{2^3}{3!} (\cos 3x + i \sin 3x) + \dots \end{aligned}$$

so $C + iS = 1 + i + 2z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$

move to remove

where $z = \cos x + i \sin x$

$$C + iS - i = 1 + 2z + \frac{(2z)^2}{2!} + \dots$$

✓ macLaurin's for e^{2z}

$$= e^{2z}$$

so

$$C + iS - i = e^{2 \cos x + 2i \sin x}$$

$$= e^{2 \cos x} e^{2i \sin x}$$

$$= e^{2 \cos x} (\cos(2 \sin x) + i \sin(2 \sin x))$$

so

$$C = e^{2 \cos x} \cos(2 \sin x)$$

$$S - 1 = e^{2 \cos x} \sin(2 \sin x)$$

$$S = e^{2 \cos x} \sin(2 \sin x) + 1$$

b) The other series

$$1 + 2\cos\theta + \frac{2^2}{2!}\cos 2\theta + \dots = e^{2\cos x} \cos(2\sin x)$$

